

ISE 331: Fundamentals of Computer Security

Practice Problems for Lectures 30 - 31 (Hardware and AI)

Directions: Below is a set of exercises and problems you should do for practice. Answering this set of questions should be just one part of your preparation for quizzes and exams. In addition to solving these problems you should:

- Review the lecture notes.
- Review your homework assignments.
- Form a study group and test each other on the material.

Note: The solution is provided on the last page. It is strongly recommended that you work on the questions first before going to the solution.

Part I. Multiple Choice

1. In hardware verification, the gap between when a device is inspected and when it is actually deployed exposes a fundamental vulnerability. This vulnerability is best described as:
 - A. A physical fault injection vulnerability.
 - B. An algorithm-level design failure.
 - C. A Time-of-Check-to-Time-of-Use (TOCTOU) problem.
 - D. A passive electromagnetic side-channel emission.
2. About one million chips are imported from a foreign country and need inspection. Which option below is NOT a valid reason why chip inspection is very difficult?
 - A. Modern chips are very small and cannot be inspected with the naked eye.
 - B. Inspection of one chip does not guarantee the safety of the other chips.
 - C. Even with proper devices, some malicious modifications can be overlooked.
 - D. Inspection of a chip cannot be performed without completely destroying it.
3. An attacker wants to extract a secret key from a smart card. The attacker chooses to monitor power variations while the card encrypts a message. This attack is classified as:
 - A. Active Direct Tampering
 - B. Side-channel Analysis
 - C. Active Fault Injection
 - D. Supply Chain Tampering
4. An attacker places carefully designed, semi-transparent stickers on a car license plate, causing an AI-based license plate recognition system to misidentify the numbers. Humans looking at the plate can still easily read the original numbers. This is an example of a(n):
 - A. Data poisoning attack
 - B. Evasion attack
 - C. Model stealing attack
 - D. Membership inference attack
5. A company released a new Large Language Model (LLM). Its adversaries query this model millions of times with selected inputs and save the corresponding outputs. This is most likely to be _____.
 - A. Model Stealing (Distillation)
 - B. Evasion Attack
 - C. Membership Inference
 - D. Spamming Attack

6. Which statement best describes the relationship between a machine learning model's accuracy on test data and its robustness against adversarial examples?
- A. A model with lower accuracy will be more robust against adversarial examples.
 - B. A model with higher accuracy will be more robust against adversarial examples.
 - C. A model can have high accuracy but remain vulnerable to adversarial examples.
 - D. A model is invulnerable to adversarial examples once its accuracy passes a threshold.

Part II. True or False

- 1. Supply chain tampering can be solved by open chip design, open PCB design, and open testing.
- 2. To ease the process of hardware verification, it is generally necessary to sacrifice features and usability.
- 3. Tamper-evident mechanisms are designed to prevent or stop an attacker from direct physical tampering.
- 4. The machine learning assumption of I.I.D. (Independent and Identically Distributed) data holds true and applies with adversarial AI security settings.
- 5. Data poisoning can happen to both discriminative models and generative models.
- 6. An attacker needs to know part of a model's internal parameters to start model stealing attack.

Part III. Short Answer

- 1. It is easy to verify the authenticity of a software installation package by checking its hash code against the hash published on the vendor's website. Why can't you do the same for hardware devices or computer chips?
- 2. Explain the core difference between the goals of Data Poisoning Attacks versus Membership Inference Attacks. Which part(s) of the CIA triad (Confidentiality, Integrity, Availability) is/are compromised if each attack is successfully performed?

Part I Solution:
1-5 CDBBA 6 C

Part II Solution:
1-5 FTFFT 6 F

Part III Solution:

1. You cannot “hash” hardware. For a software package, it is fundamentally 0s and 1s, and can be processed by hash algorithms relatively easily. There is no easy way to do the same for hardware.
2. **Data poisoning attack:** The goal is to compromise the performance of the model, so it makes incorrect predictions or generate unsatisfactory outputs. Hurts the integrity/availability of the model.

Membership inference attack: The goal is to determine whether a specific piece of data belongs to the training dataset or not. Hurts the confidentiality of the model.